

LAZY LEARNING

Lazy Learning

- **Lazy Learning** - methods simply store the data and **generalizing** beyond these **data** is **postponed** until an **explicit request** is made.
- eg:- Nearest neighbor classifiers

Understanding Nearest Neighbor Classification

- **Nearest neighbor** classifiers are defined by their characteristic of classifying unlabeled examples by assigning them the class of similar labeled examples.
- Nearest neighbor classifiers are well-suited for classification tasks, where relationships among the features and the target classes are numerous, complicated, or extremely difficult to understand, yet the items of similar class type tend to be fairly homogeneous.
- if the data is noisy and thus no clear distinction exists among the groups, the nearest neighbor algorithms may struggle to identify the class boundaries.

Eg:

- Computer vision applications, including optical character recognition and facial recognition in both still images and video
- Predicting whether a person will enjoy a movie or music recommendation
- Identifying patterns in genetic data, perhaps to use them in detecting specific proteins or diseases

The k-NN algorithm

- The nearest neighbor's approach to classification is exemplified by the **k-nearest neighbor's algorithm (k-NN)**.
- One of the simplest machine learning algorithms.
- It is still used widely.

The strengths and weaknesses of this algorithm are as follows:

STRENGTH	WEAKNESS
• Simple and effective	• Does not produce a model, limiting the ability to understand how the features are related to the class
• Makes no assumptions about the underlying data distribution	• Requires selection of an appropriate k
• Fast training phase	• Slow classification phase
	• Nominal features and missing data require additional processing

- The k -NN algorithm gets its name from the fact that it uses information about an example's k -nearest neighbors to classify unlabeled examples.
- k is a variable term implying that any number of nearest neighbors could be used.
- choosing k , the algorithm requires a training dataset made up of examples that have been classified into several categories, as labeled by a nominal variable.
- Then, for each unlabeled record in the test dataset, k -NN identifies k records in the training data that are the "nearest" in similarity.

Eg;-BlindTastingExperience

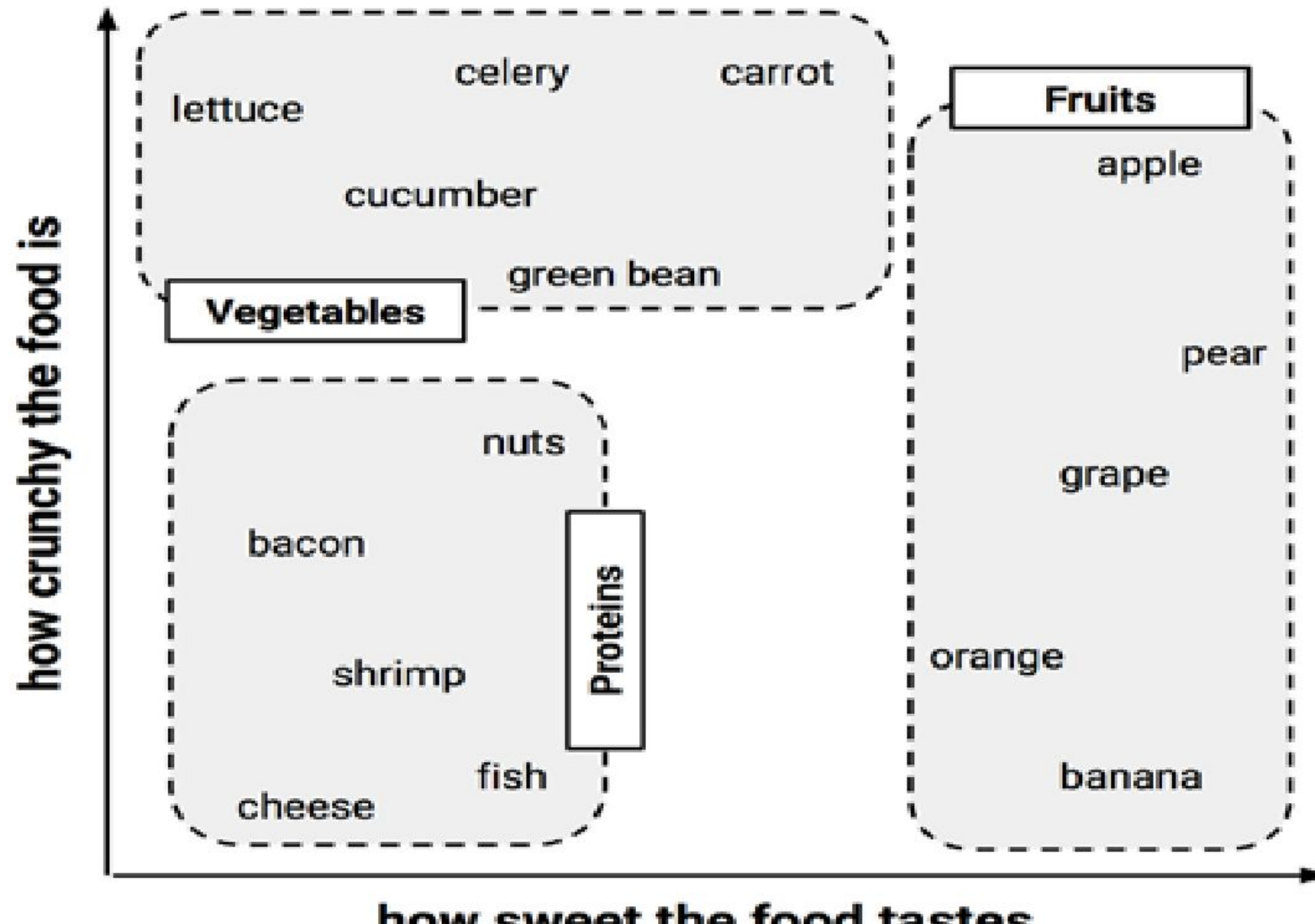
To keep things simple, we rated only two features of each ingredient. The first is a measure from 1 to 10 of how crunchy the ingredient is and the second is a 1 to 10 score of how sweet the ingredient tastes. We then labeled each ingredient as one of the three types of food: fruits, vegetables, or proteins. The first few rows of such a dataset might be structured as follows

Ingredient	Sweetness	Crunchiness	Food type
apple	10	9	fruit
bacon	1	4	protein
banana	10	1	fruit
carrot	7	10	vegetable
celery	3	10	vegetable
cheese	1	1	protein

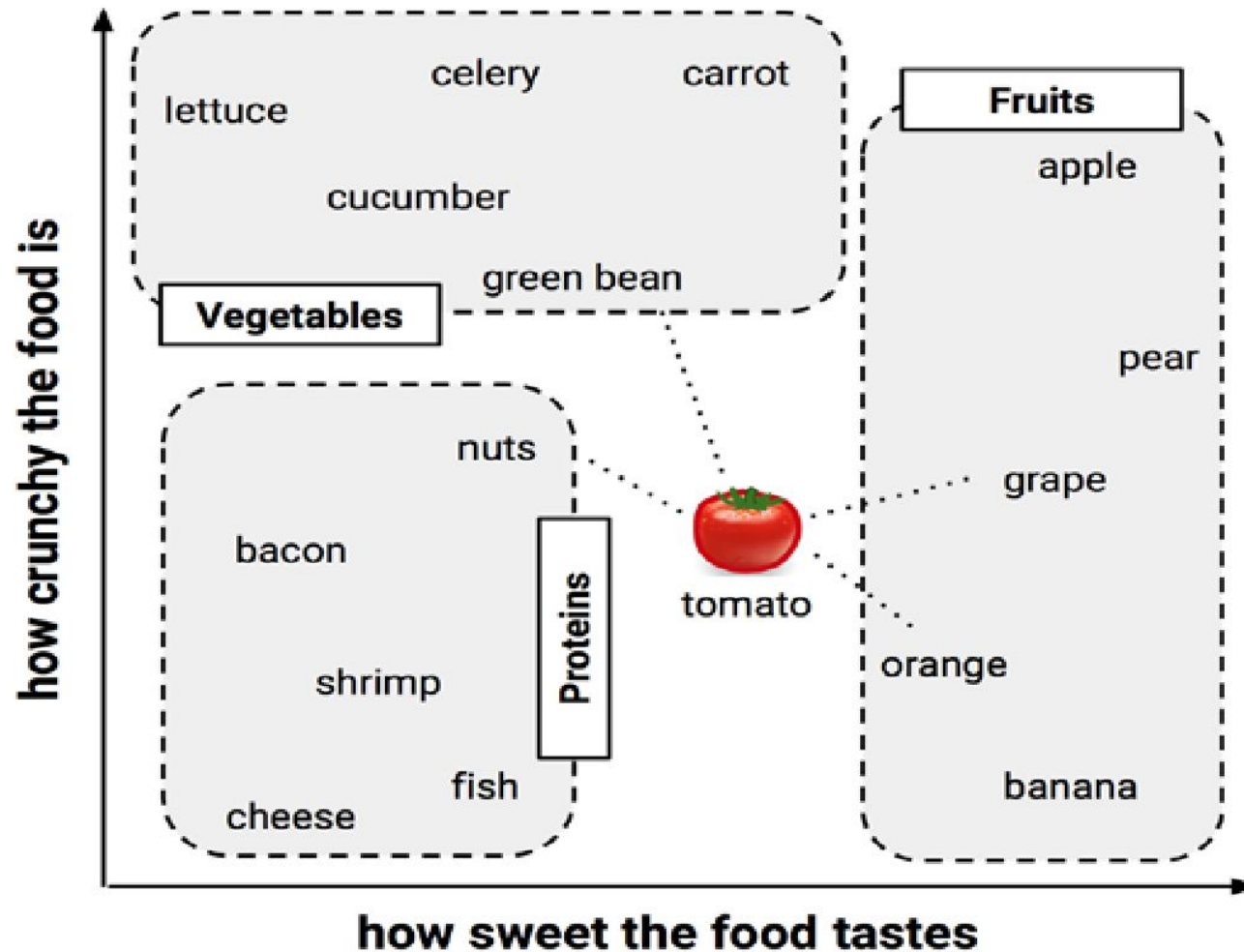
- The k-NN algorithm treats the features as coordinates in a multidimensional feature space.
- As our dataset includes only two features, the feature space is two-dimensional.
- We can plot two-dimensional data on a scatter plot, with the x dimension indicating the ingredient's sweetness and the y dimension, the crunchiness.



- Similar types of food tend to be grouped closely together.



- Suppose that after constructing this dataset, we decide to use it to settle the age-old question: is tomato a fruit or vegetable? We can use the nearest neighbor approach to determine which class is a better fit, as shown in the following diagram:



Measuring similarity with distance

- Locating the tomato's nearest neighbors requires a **distance function**, or a formula that measures the similarity between the two instances.
- There are many different ways to calculate distance.
- The k-NN algorithm uses **Euclidean distance**, which is the distance one would measure if it were possible to use a ruler to connect two points.

-illustrated in the previous figure by the dotted lines connecting the tomato to its neighbors.

-Euclidean distance is specified by the following formula:

- p and q are the examples to be compared, each having n features.
- The term p_1 refers to the value of the first feature of example p .
- while q_1 refers to the value of the first feature of example q .

$$\text{dist}(p, q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2 + \dots + (p_n - q_n)^2}$$

- The distance formula involves comparing the values of each feature.
- For example, to calculate the distance between the tomato (*sweetness* = 6, *crunchiness* = 4), and the green bean (*sweetness* = 3, *crunchiness* = 7), we can use the formula as follows:
 - **p & q** □ **examples to be compared**, each having **n - features**.
 - **p₁** □ **the value of the first feature of example p**;
 - **q₁** □ **the value of the first feature of example q**:

$$\text{dist}(p, q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2 + \dots + (p_n - q_n)^2}$$

$$\text{dist}(\text{tomato}, \text{green bean}) = \sqrt{(6 - 3)^2 + (4 - 7)^2} = 4.2$$

- In a similar vein, we can calculate the distance between the tomato and several of its closest neighbors as follows:

Ingredient	Sweetness	Crunchiness	Food type	Distance to the tomato
grape	8	5	fruit	$\text{sqrt}((6 - 8)^2 + (4 - 5)^2) = 2.2$
green bean	3	7	vegetable	$\text{sqrt}((6 - 3)^2 + (4 - 7)^2) = 4.2$
nuts	3	6	protein	$\text{sqrt}((6 - 3)^2 + (4 - 6)^2) = 3.6$
orange	7	3	fruit	$\text{sqrt}((6 - 7)^2 + (4 - 3)^2) = 1.4$

1-NN classification

- To classify the tomato as a vegetable, protein, or fruit, we'll begin by assigning the tomato, the food type of its single nearest neighbor.
- This is called 1-NN classification because $k = 1$.
- The orange is the nearest neighbor to the tomato, with a distance of 1.4.
- As orange is a fruit, the 1-NN algorithm would classify tomato as a fruit.
- If we use the k-NN algorithm with $k = 3$ instead,
Three nearest neighbors: orange, grape, and nuts.
- Since the majority class among these neighbors is fruit (two of the three votes), the tomato again is classified as a fruit.

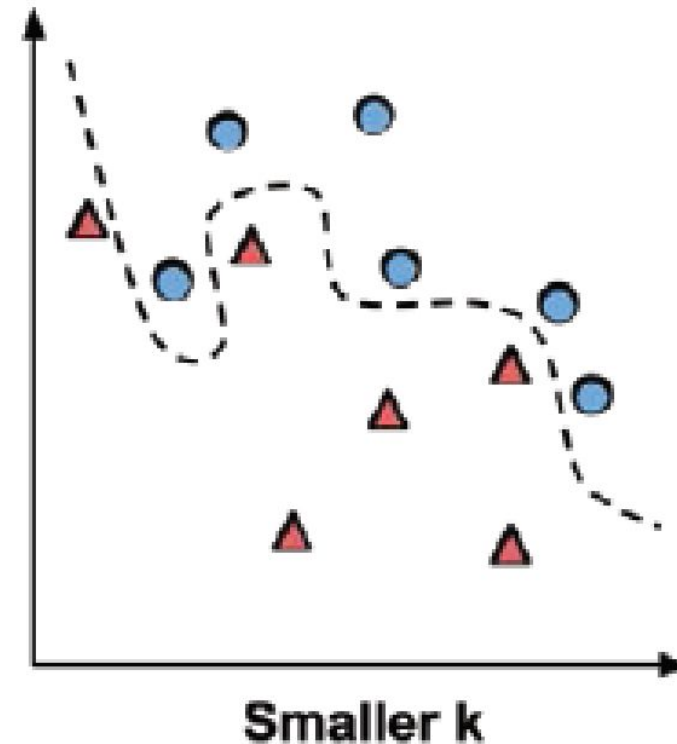
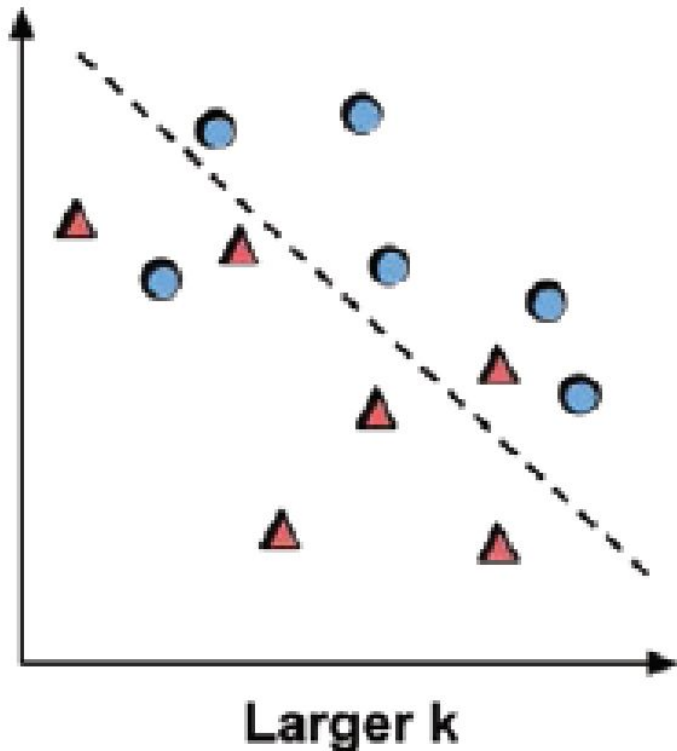
Choosing an appropriate k

- The decision of how many neighbors to use for k-NN determines how well the model will generalize to future data.
- The balance between overfitting and underfitting the training data is a problem known as **bias-variance tradeoff**.
 - **Overfitting** □ occurs when a statistical model or machine learning algorithm **captures the noise** of the data.
 - **Underfitting** □ refers to a model that can **neither model the training data nor generalize** to new data

- Choosing a large k reduces the impact or variance caused by noisy data, but can bias the learner so that it runs the risk of ignoring small, but important patterns.
- The extreme stance of setting a very large k , as large as the total number of observations in the training data.
- With every training instance represented in the final vote, the most common class always has a majority of the voters.

- The model would consequently always predict the majority class, regardless of the nearest neighbors.
- Single nearest neighbor allows the noisy data or outliers to unduly influence the classification of examples.
- The best k value is somewhere between these two extremes.

- The following figure illustrates, more generally, how the decision boundary (depicted by a dashed line) is affected by larger or smaller k values.
- Smaller value allow more complex decision boundaries that more carefully fit the training data.
- The problem is that we do not know whether the straight boundary or the curved boundary better represents the true underlying concept to be learned.



- To begin with $k = \text{square root of the number of training examples}$.
- In the food classifier we developed previously, we might set $k = 4$ because there were $15 \text{ example ingredients}$ in the training data and
- square root of $15 = 3.87$.

- Based on the survey conducted in an institution the students are classified based on the 2 attributes academic excellence and other achievements. Consider the data set given. Find the classification of a student with value of X is 5 and Y is 7 based on the data of trained samples using KNN algorithm. Choose $k = 3$

X [Academic Excellence]	Y [Activities]	Z [Classification]
8	6	Outstanding
5	6	Good
7	3	Good
6	9	Outstanding

Preparing data for use with k-NN

- Features are typically transformed to a standard range prior to applying the k-NN algorithm.
- The rationale for this step is that the distance formula is highly dependent on how features are measured.
- In particular, if certain features have a much larger range of values than the others, the distance measurements will be strongly dominated by the features with larger ranges.
- The solution is to rescale the features by shrinking or expanding their range such that each one contributes relatively equally to the distance formula.

- For example, if sweetness and crunchiness are both measured on a scale from 1 to 10, we would also like spiciness to be measured on a scale from 1 to 10. There are several common ways to accomplish such scaling.
- **There are several methods for scaling.**
 - **min-max** normalization.
 - **z-score** standardization.
 - **dummy coding.**

Min-max normalization

- The traditional method of rescaling features for k-NN is **min-max normalization**.
- This process transforms a feature such that all of its values fall in a range between 0 and 1.
- The formula for normalizing a feature is as follows:

$$X_{new} = \frac{X - \min(X)}{\max(X) - \min(X)}$$

- Essentially, the formula subtracts the minimum of feature X from each value and divides by the range of X .

z-score standardization.

- Another common transformation is called **z-score standardization**.
- The following formula subtracts the mean value of feature X , and divides the outcome by the standard deviation of X :
- Rescales each of the feature's values in terms of how many standard deviations they fall above or below the mean value.

$$X_{new} = \frac{X - \mu}{\sigma} = \frac{X - \text{Mean}(X)}{\text{StdDev}(X)}$$

- The resulting value is called a **z-score**.
- The z-scores fall in an unbound range of negative and positive numbers.
- Unlike the normalized values, they have no predefined minimum and maximum.

Dummy coding

- The Euclidean distance formula is not defined for nominal data.
- Therefore, to calculate the distance between nominal features, we need to convert them into a numeric format.
- A typical solution utilizes **dummy coding**, where a value of 1 indicates one category, and 0 , the other.
- For instance, dummy coding for a gender variable could be constructed as:

$$\text{male} = \begin{cases} 1 & \text{if } x = \text{male} \\ 0 & \text{otherwise} \end{cases}$$

- Dummy coding of the two-category (binary) gender variable results in a single new feature named male.
- There is no need to construct a separate feature for female.
- Since the two sexes are mutually exclusive, knowing one or the other is enough.
- An n -category nominal feature can be dummy coded by creating the binary indicator variables for $(n - 1)$ levels of the feature.

-For example, the dummy coding for a three-category temperature variable (for example, hot, medium, or cold) could be set up as $(3 - 1) = 2$ features, as shown here:

- -For example, the dummy coding for a three-category temperature variable (for example, hot, medium, or cold) could be set up as $(3 - 1) = 2$ features, as shown here:

$$\text{hot} = \begin{cases} 1 & \text{if } x = \text{hot} \\ 0 & \text{otherwise} \end{cases}$$
$$\text{medium} = \begin{cases} 1 & \text{if } x = \text{medium} \\ 0 & \text{otherwise} \end{cases}$$

- Knowing that hot and medium are both 0 is enough to know that the temperature is cold.
- Therefore, do not need a third feature for the cold category.
- A convenient aspect of dummy coding is that the distance between dummy coded features is always one or zero, and thus, the values fall on the same scale as min-max normalized numeric data.
- No additional transformation is necessary.

Why is the k-NN algorithm lazy?

- No abstraction occurs.
- The abstraction and generalization processes are skipped altogether, and this undermines the definition of learning.
- A lazy learner is not really learning anything.
- It merely stores the training data verbatim.
- This allows the training phase, which is not actually training anything, to occur very rapidly.
- The downside is that the process of making predictions tends to be relatively slow in comparison to training.
- Lazy learning is also known as **instance-based learning** or **rote learning**.

Probabilistic Learning

- When a meteorologist provides a weather forecast, precipitation is typically described with terms such as "70 percent chance of rain."
- Such forecasts are known as probability of precipitation reports..
- Weather estimates are based on probabilistic methods or those concerned with describing uncertainty.
- They use data on past events to extrapolate future events. In the case of weather, the chance of rain describes the proportion of prior days to similar measurable atmospheric conditions in which precipitation occurred. A 70 percent chance of rain implies that in 7 out of the 10 past cases with similar conditions, precipitation occurred somewhere in the area.

Understanding Naive Bayes

- The technique descended from the work of the 18th century mathematician Thomas Bayes,
- Who developed foundational principles to describe the probability of events, and how probabilities should be revised in the light of additional information.
- These principles formed the foundation for what are now known as **Bayesian methods**.

- probability - number between 0 and 1 (that is, between 0 % and 100 %)
- which captures the chance that an event will occur in the light of the available evidence.
- lower probability - less likely the event is to occur.
- probability of 0 - event will definitely not occur
- probability of 1 - that the event will occur with 100 percent certainty.

- Classifiers based on Bayesian methods utilize training data to calculate an observed probability of each outcome based on the evidence provided by feature values.
- When the classifier is later applied to unlabeled data, it uses the observed probabilities to predict the most likely class for the new features.
- It's a simple idea, but it results in a method that often has results on equal with more sophisticated algorithms.
- Bayesian classifiers have been used for:
 - Text classification, such as junk e-mail (spam) filtering.
 - Intrusion or anomaly detection in computer networks.
 - Diagnosing medical conditions given a set of observed symptoms.

- Bayesian classifiers are best applied to problems in which the information from numerous attributes should be considered simultaneously in order to estimate the overall probability of an outcome.
- many machine learning algorithms ignore features that have weak effects, Bayesian methods utilize all the available evidence to subtly change the predictions.
- If large number of features have relatively minor effects, taken together, their combined impact could be quite large.

Basic concepts of Bayesian methods

- Bayesian probability theory is rooted in the idea that the estimated likelihood of an **event**, or a potential outcome, should be based on the evidence at hand across multiple **trials**, or opportunities for the event to occur.

- The following table illustrates events and trials for several real-world outcomes:

Event	Trial
Heads result	Coin flip
Rainy weather	A single day
Message is spam	Incoming e-mail message
Candidate becomes president	Presidential election
Win the lottery	Lottery ticket

Understanding probability

- The probability of an event is estimated from the observed data by dividing the number of trials in which the event occurred by the total number of trials.
- **Probability = (Number of trials in which the event occurred / Total number of trials.)**
- Eg:- if it rained 3 out of 10 days with similar conditions as today, the probability of rain today = $3 / 10 = 0.30$ % or 30 percent.
- if 10 out of 50 prior email messages were spam, then the probability of any incoming message being spam = $10 / 50 = 0.20\%$ or 20 percent.

- $P(A)$ - probability of event A .

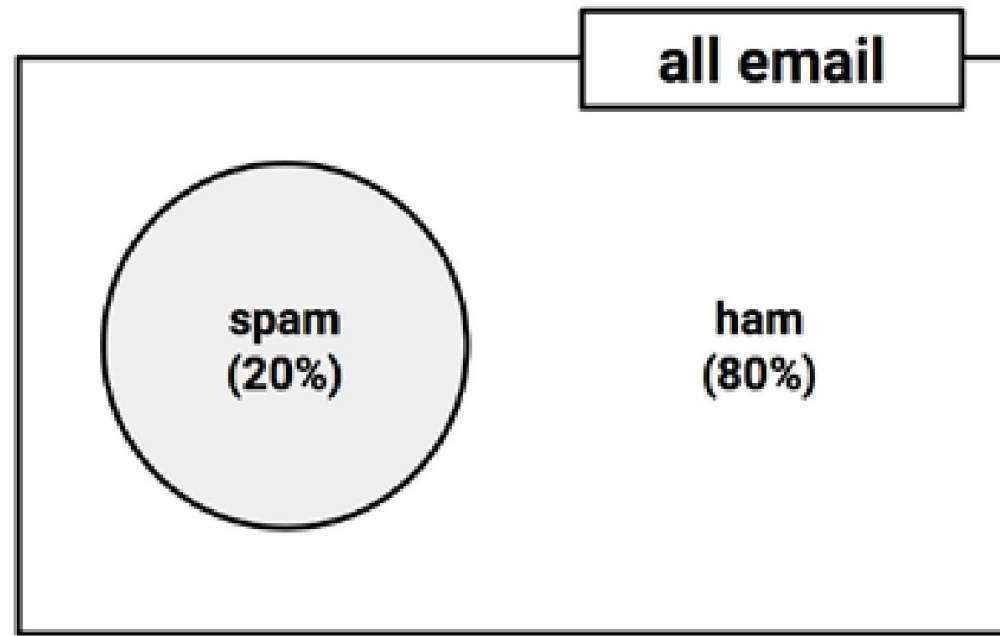
Eg:- $P(\text{rain}) = 0.30$

$P(\text{spam}) = 0.20$.

- probability of all the possible outcomes of a trial must always sum to 1, because a trial always results in some outcome happening.
- an event is always mutually exclusive and exhaustive with its **complement**.

- The complement of event A is typically denoted A^c or A' .
- $P(\neg A)$ can be used to denote the probability of event A not occurring.
- **Eg: - $P(\text{spam}) = 0.20$.**
 - $P(\neg \text{ham}) = 1 - 0.20 = 0.80$.
- This notation is equivalent to $P(A^c)$.
- This concludes that spam and ham are **mutually exclusive and exhaustive** events,

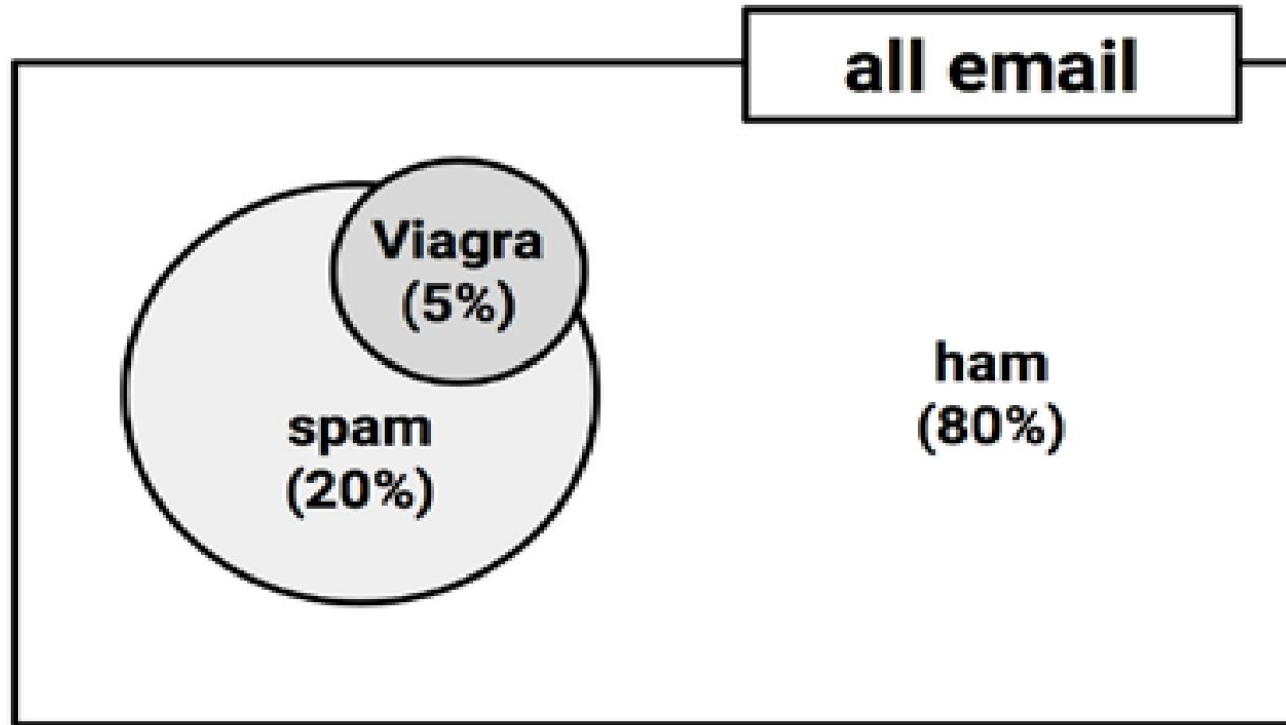
- Eg: The circle represents the 20 percent probability that the message is spam. The remaining 80 percent represents the complement $P(\neg spam)$ or the messages that are not spam:(two-dimensional space that is partitioned into probabilities for each event,the rectangle represents the possible outcomes for an e-mail message.)



Understanding joint probability

- If certain events occur with the event of interest, we may be able to use them to make predictions.
- Eg: Consider, second event based on the outcome that an e-mail message contains the word “*Viagra.*”(In most cases, this word is likely to appear only in a spam message)
- its presence in an incoming e-mail is therefore a very strong piece of evidence that the message is spam.

- The preceding diagram, updated for this second event, might appear as shown in the following diagram:

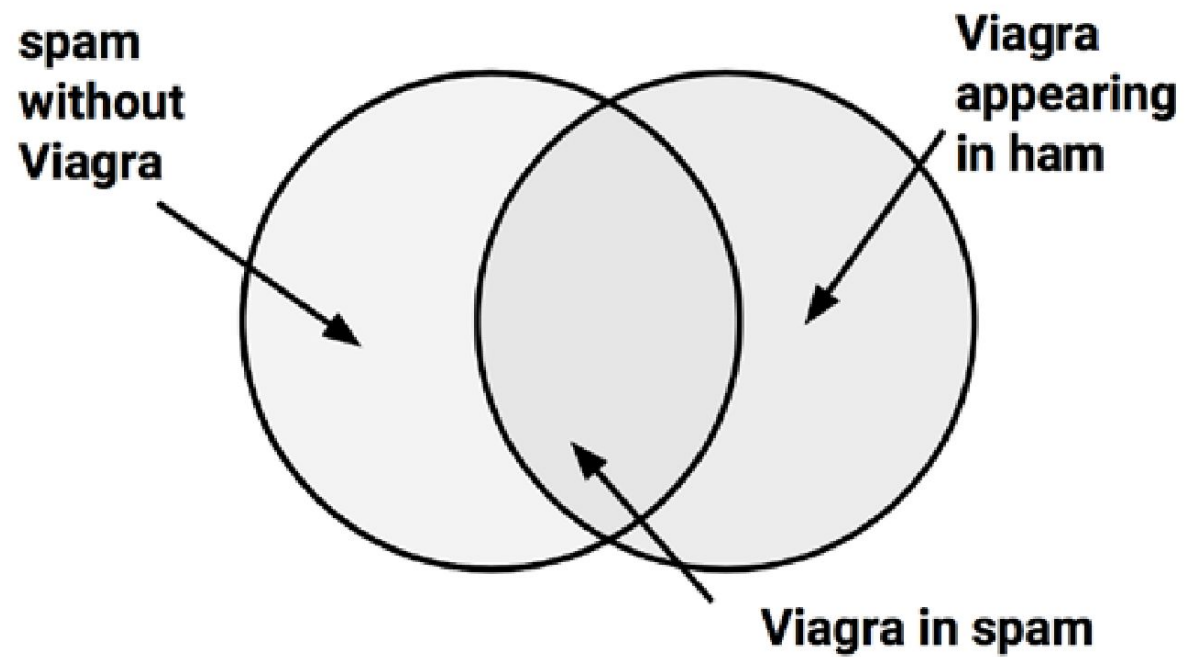


- This implies that not all spam messages contain the word Viagra and not every e-mail with the word Viagra is spam.

Venn diagram.

- To zoom in for a closer look at the overlap between the spam and Viagra circles, we'll employ a visualization known as a **Venn diagram**.
- First used in the late 19th century ,by **John Venn**.
- The diagram uses circles to illustrate the overlap between sets of items.
- It is used as a reminder to allocate probability to all possible combinations of events.

- Venn diagram



- the probability that both $P(spam)$ and $P(Viagra)$ occur, which can be written as $P(spam \cap Viagra)$.
- Calculating $P(spam \cap Viagra)$ depends on the **joint probability** of the two events or how the probability of one event is related to the probability of the other.
- If the two events are totally unrelated, they are called **independent events**.
 - This is not to say that independent events cannot occur at the same time; event independence simply implies that knowing the outcome of one event does not provide any information about the outcome of the other.

- If all events were independent, it would be impossible to predict one event by observing another. In other words, **dependent events** are the basis of predictive modeling.
- the presence of clouds is predictive of a rainy day, the appearance of the word Viagra is predictive of a spam e-mail.
- Calculating the probability of dependent events is a bit more complex than for independent events.
- If $P(\text{spam})$ and $P(\text{Viagra})$ were independent, we could easily calculate $P(\text{spam} \cap \text{Viagra})$, the probability of both events happening at the same time.

- Because 20 percent of all the messages are spam, and 5 percent of all the e-mails contain the word Viagra, we could assume that 1 percent of all messages are spam with the term Viagra. This is because $0.05 * 0.20 = 0.01$. More generally, for independent events A and B , the probability of both happening can be expressed as $P(A \cap B) = P(A) * P(B)$.

- independent events A and B , the probability of both happening can be expressed as

$$P(A \cap B) = P(A) * P(B).$$

- $P(A) = 20\% = 0.2$
- $P(B) = 5\% = 0.05$;
- $P(A \cap B) = 0.05 * 0.20 = 0.01$;
- $P(spam)$ and $P(Viagra)$ are likely to be highly dependent, which means that this calculation is incorrect.
- To obtain a reasonable estimate, we need to use a more careful formulation of the relationship between these two events, which is based on advanced Bayesian methods.

Computing conditional probability with Bayes' theorem

- The relationships between dependent events can be described using **Bayes' theorem**
- Estimate of the probability of one event in light of the evidence provided by another event:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- The notation $P(A|B)$ is read as the probability of event A , given that event B occurred. This is known as **conditional probability**.
- Since the probability of A is dependent (that is, conditional) on what happened with event B .

- Bayes' theorem tells that estimate of $P(A|B)$ should be based on ;
- $-P(A \cap B)$, a measure of how often A and B are observed to occur together,
- $-P(B)$, a measure of how often B is observed to occur in general

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- **$P(A \cap B) = P(A|B) * P(B)$** , a fact that can be easily derived by applying a bit of algebra to the previous formula.
- Rearranging this formula once more with the knowledge that :
 $P(A \cap B) = P(B \cap A)$
- **So, $P(A \cap B) = P(B|A) * P(A)$, use this in Bayes theorem: we get;**
- which we can then use in the following formulation of Bayes' theorem:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

Prior Probability:

- The prior probability of an event is the probability of the event computed before the collection of new data.
- Without knowledge of an incoming message's content, the best estimate of its spam status would be $P(spam)$, the probability that any prior message was spam, which we calculated previously to be 20 percent.
- This estimate is known as the **prior probability**.

Likelihood:

- The probability that the word Viagra was used in previous spam messages, or $P(\text{Viagra}|\text{spam})$, is called the **likelihood**.

Marginal likelihood

- The probability that Viagra appeared in any message at all, or $P(\text{Viagra})$, is known as the **marginal likelihood**.

Posterior Probability :

- Probability of event A occurring given that event B has occurred.
- Calculated by updating the prior probability by using Bayes' theorem.
- we can compute a posterior probability that measures how likely the message is to be spam.
- *Eg: $P(\text{spam}|\text{viagra})$* - how likely the message is to be spam.

- If the posterior probability is greater than 50 percent, the message is more likely to be spam than ham and it should perhaps be filtered.
- The following formula shows how Bayes' theorem is applied to the evidence provided by the previous e-mail messages:

$$P(\text{spam}|\text{Congrats}) = \frac{P(\text{Congrats}|\text{spam})P(\text{spam})}{P(\text{Congrats})}$$

Diagram illustrating the components of Bayes' theorem formula:

- $P(\text{spam}|\text{Congrats})$ is labeled as the **posterior probability**.
- $P(\text{Congrats}|\text{spam})$ is labeled as the **likelihood**.
- $P(\text{spam})$ is labeled as the **prior probability**.
- $P(\text{Congrats})$ is labeled as the **marginal likelihood**.

- To calculate these components of Bayes' theorem, it helps to construct a **frequency table**, that records the number of times Viagra appeared in spam and ham messages.

	Viagra		
Frequency	Yes	No	Total
spam	4	16	20
ham	1	79	80
Total	5	95	100

	Viagra		
Likelihood	Yes	No	Total
spam	$4 / 20$	$16 / 20$	20
ham	$1 / 80$	$79 / 80$	80
Total	$5 / 100$	$95 / 100$	100

- The likelihood table reveals that $P(\text{Viagra}=\text{Yes}|\text{spam}) = 4/20 = 0.20$, indicating that the probability is 20 percent that a message contains the term Viagra, given that the message is spam.
- Additionally, since $P(A \cap B) = P(B|A) * P(A)$, we can calculate
- $P(\text{spam} \cap \text{Viagra})$ as $P(\text{Viagra}|\text{spam}) * P(\text{spam})$
 $= (4/20) * (20/100) = 0.04$.
- The same result can be found in the frequency table, which notes that 4 out of the 100 messages were spam with the term Viagra.

- To compute the posterior probability, $P(\text{spam}|\text{Viagra})$, we simply take
- $P(\text{Viagra}|\text{spam}) * P(\text{spam}) / P(\text{Viagra})$
 $(4/20) * (20/100) / (5/100) = 0.80$.
- Therefore, the probability is 80 percent that a message is spam, given that it contains the word Viagra.
- In light of this result, any message containing this term should probably be filtered.
- This is very much how commercial spam filters work, although they consider a much larger number of words simultaneously while computing the frequency and likelihood tables.

The Naive Bayes algorithm

- The **Naive Bayes** algorithm describes a simple method to apply Bayes' theorem to classification problems.
- Although it is not the only machine learning method that utilizes Bayesian methods, it is the most common one.
- This is particularly true for text classification, where it has become the de facto standard.

The strengths and weaknesses of this algorithm

Strengths	Weaknesses
• Simple, fast, and very effective • Does well with noisy and missing data • Requires relatively few examples for training, but also works well with very large numbers of examples • Easy to obtain the estimated probability for a prediction	• Relies on an often-faulty assumption of equally important and independent features • Not ideal for datasets with many numeric features • Estimated probabilities are less reliable than the predicted classes

- The Naive Bayes algorithm is named as such because it makes some "naive" assumptions about the data.
- In particular, Naive Bayes assumes that all of the features in the dataset are equally important and independent.
- These assumptions are rarely true in most real-world applications.

Eg:

- If you were attempting to identify spam by monitoring e-mail messages, it is almost certainly true that some features will be more important than others.
- The e-mail sender may be a more important indicator of spam than the message text.
- Additionally, the words in the message body are not independent from one another, since the appearance of some words is a very good indication that other words are also likely to appear.
- A message with the word Viagra will probably also contain the words prescription or drugs.

- In most cases when these assumptions are violated, Naive Bayes still performs fairly well. This is true even in extreme circumstances where strong dependencies are found among the features.
- Due to the algorithm's versatility and accuracy across many types of conditions, Naive Bayes is often a strong first candidate for classification learning tasks.

Classification with Naive Bayes

Eg: The Naive Bayes learner is trained by constructing a likelihood table for the appearance of these four words (labeled W_1 , W_2 , W_3 , and W_4), as shown in the following diagram for 100 e-mails:

	Viagra (W_1)		Money (W_2)		Groceries (W_3)		Unsubscribe (W_4)		
Likelihood	Yes	No	Yes	No	Yes	No	Yes	No	Total
spam	4 / 20	16 / 20	10 / 20	10 / 20	0 / 20	20 / 20	12 / 20	8 / 20	20
ham	1 / 80	79 / 80	14 / 80	66 / 80	8 / 80	71 / 80	23 / 80	57 / 80	80
Total	5 / 100	95 / 100	24 / 100	76 / 100	8 / 100	91 / 100	35 / 100	65 / 100	100

- we can define the problem as shown in the following formula.
- It captures the probability that a message is spam, given that *Viagra* = *Yes*, *Money* = *No*, *Groceries* = *No*, and *Unsubscribe* = *Yes*:

$$P(\text{spam} | W_1 \cap \neg W_2 \cap \neg W_3 \cap W_4) = \frac{P(W_1 \cap \neg W_2 \cap \neg W_3 \cap W_4 | \text{spam}) P(\text{spam})}{P(W_1 \cap \neg W_2 \cap \neg W_3 \cap W_4)}$$

- This formula is computationally difficult to solve.
- As additional features are added, tremendous amounts of memory are needed to store probabilities for all of the possible intersecting events; imagine the complexity of a Venn diagram for the events for four words, let alone for hundreds or more.

- The work becomes much easier if we can exploit the fact that Naive Bayes assumes independence among events.
- Specifically, it assumes **class-conditional independence**, which means that events are independent so long as they are conditioned on the same class value.
- Assuming conditional independence allows us to simplify the formula using the probability rule for independent events, which states that

$$P(A \cap B) = P(A) * P(B)$$

- The conditional probability of spam can be expressed as:

$$P(\text{spam}|W_1 \cap \neg W_2 \cap \neg W_3 \cap W_4) \propto P(W_1|\text{spam})P(\neg W_2|\text{spam})P(\neg W_3|\text{spam})P(W_4|\text{spam})P(\text{spam})$$

- The probability that the message is ham can be expressed as:

$$P(\text{ham}|W_1 \cap \neg W_2 \cap \neg W_3 \cap W_4) \propto P(W_1|\text{ham})P(\neg W_2|\text{ham})P(\neg W_3|\text{ham})P(W_4|\text{ham})P(\text{ham})$$

- the equals symbol has been replaced by the proportional-to symbol (similar to a sideways, open-ended '8') to indicate the fact that the denominator has been omitted.
- Using the values in the likelihood table, we can start filling numbers in these equations. The overall likelihood of spam is then:
$$(4/20) * (10/20) * (20/20) * (12/20) * (20/100) = 0.012$$
- While the likelihood of ham is:
$$(1/80) * (66/80) * (71/80) * (23/80) * (80/100) = 0.002$$
- Because $0.012/0.002 = 6$, we can say that this message is six times more likely to be spam than ham.

- However, to convert these numbers into probabilities, we need to perform one last step to reintroduce the denominator that had been excluded.
 - Essentially, we must rescale the likelihood of each outcome by dividing it by the total likelihood across all possible outcomes.
 - **Probability(spam) = likelihood(spam) / likelihood(spam+ham);**
= 0.012 / (0.012 + 0.002) = 0.857;
- **Probability(spam) = 85.7 %**

- **Probability(ham)** = likelihood that the message is ham / likelihood that the message is either spam or ham:

$$= 0.002 / (0.012 + 0.002) = \mathbf{0.143}$$

- **Probability(ham) = 14.3%.**
- Given the pattern of words found in this message, we expect that the message is spam with 85.7 percent probability and ham with 14.3 percent probability.
- Because these are mutually exclusive and exhaustive events, the probabilities sum to 1.

- The Naive Bayes classification algorithm we used in the preceding example can be summarized by the following formula.

$$P(C_L|F_1, \dots, F_n) = \frac{1}{Z}p(C_L) \prod_{i=1}^n p(F_i|C_L)$$

- **CL** - Level for class C,
- **F1,..Fn** - features F1 through Fn,
- **1/Z** - scaling factor(which converts the **likelihood** values into **probabilities**):

- Before we employ Naive Bayes in more complex problems, there are some nuances to consider.
- Suppose that we received another message, this time containing all four terms: Viagra, Groceries, Money, and Unsubscribe.
- Using the Naive Bayes algorithm as before, we can compute the likelihood of spam as:

$$(4/20) * (10/20) * (0/20) * (12/20) * (20/100) = 0$$

- The likelihood of ham is:

$$(1/80) * (14/80) * (8/80) * (23/80) * (80/100) = 0.00005$$

- Therefore, the probability of spam is:

$$0/(0 + 0.00005) = 0$$

- The probability of ham is:

$$0.00005/(0 + 0.00005) = 1$$

- These results suggest that the message is spam with 0 % probability and ham with 100 % probability.
- Does this prediction make sense? Probably not. The message contains several words usually associated with spam, including Viagra, which is rarely used in legitimate messages. It is therefore very likely that the message has been incorrectly classified. This problem might arise if an event never occurs for one or more levels of the class. For instance, the term Groceries had never previously appeared in a spam message. Consequently,

$$P(\textit{spam}|\textit{groceries}) = 0\%.$$

- Because probabilities in the Naive Bayes formula are multiplied in a chain, this 0 % value causes the posterior probability of spam to be 0, giving the word Groceries the ability to effectively nullify and overrule all of the other evidence.
- Even if the e-mail was otherwise overwhelmingly expected to be spam, the absence of the word Groceries in spam will always veto the other evidence and result in the probability of spam being 0.

- A solution to this problem involves using something called the **Laplace estimator**, which is named after the French mathematician Pierre-Simon Laplace.
- The Laplace estimator essentially adds a small number to each of the counts in the frequency table, which ensures that each feature has a nonzero probability of occurring with each class.
- Typically, the Laplace estimator is set to 1, which ensures that each class-feature combination is found in the data at least once.

- Using a Laplace value of 1, we add one to each numerator in the likelihood function.
- The total number of 1 values must also be added to each conditional probability denominator.
- The likelihood of spam is therefore:
$$(5/24) * (11/24) * (1/24) * (13/24) * (20/100) = 0.0004$$
- The likelihood of ham is:
$$(2/84) * (15/84) * (9/84) * (24/84) * (80/100) = 0.0001$$
- **Probability of spam = 80 %**
- **Probability of ham = 20 %**
- Which is a more acceptable result than the one obtained when the term Groceries alone determined the result.

Using numeric features with Naive Bayes

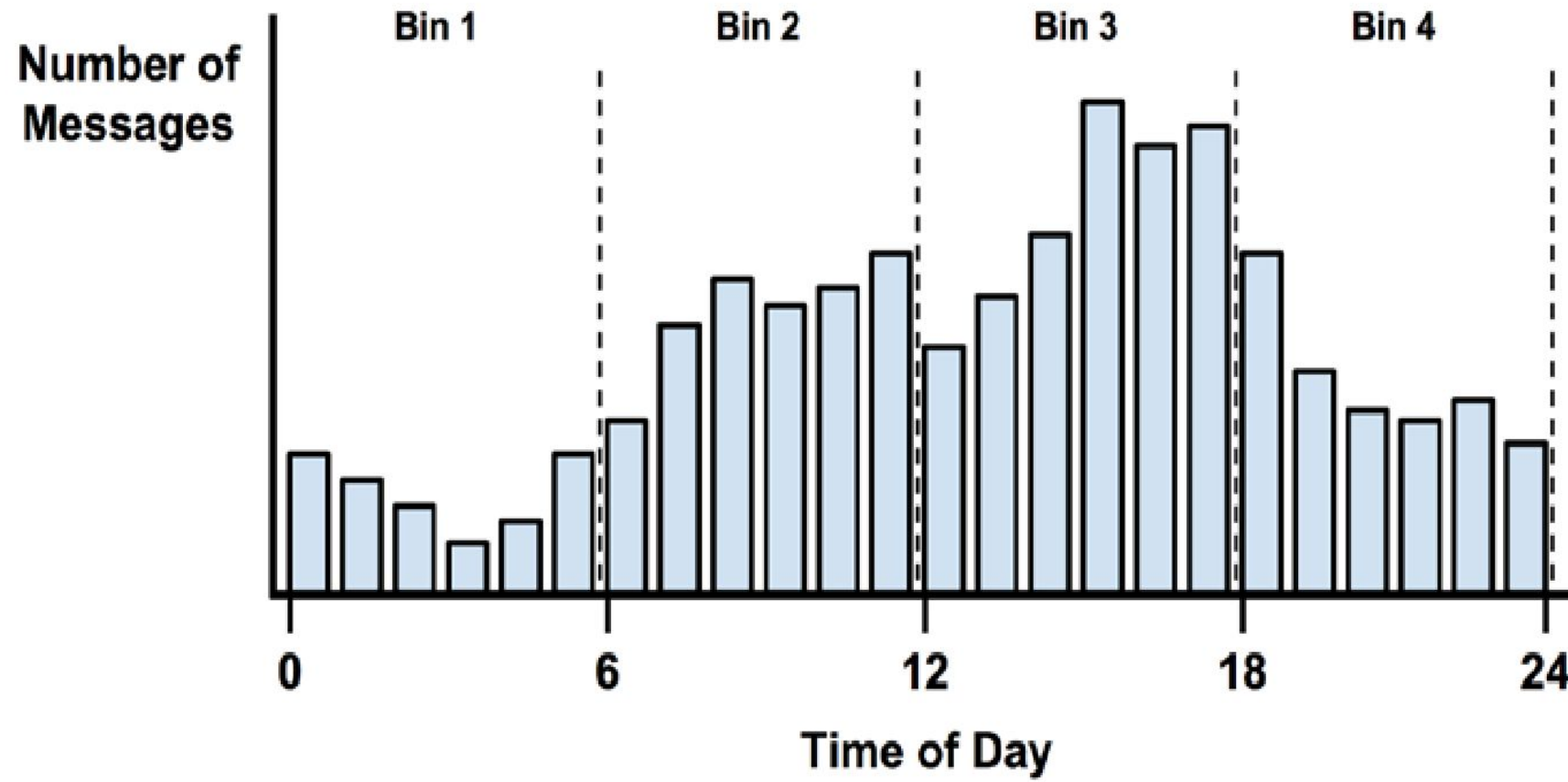
- Naive Bayes uses frequency tables to learn the data, each feature must be categorical in order to create the combinations of class and feature values comprising of the matrix.
- Since numeric features do not have categories of values, the preceding algorithm does not work directly with numeric data.

There are, however, ways that this can be addressed.

- One easy and effective solution is to **discretize** numeric features, which simply means that the numbers are put into categories known as **bins**.
- For this reason, discretization is also sometimes called **binning**.
- This method is ideal when there are large amounts of training data, a common condition while working with Naive Bayes.

- There are several different ways to discretize a numeric feature.
- Perhaps the most common is to explore the data for natural categories or **cut points** in the distribution of data.
- Eg:-, suppose that you added a feature to the spam dataset that recorded the time of night or day the e-mail was sent, from 0 to 24 hours past midnight.

- Histogram of the time data might look something like the following diagram:



- In the early hours of the morning, the message frequency is low.
- The activity picks up during business hours and tapers off in the evening.
- This seems to create four natural bins of activity, as partitioned by the dashed lines indicating places where the numeric data are divided into levels of a new nominal feature, which could then be used with Naive Bayes

Choice of four bins

- The choice of **four bins** was somewhat arbitrary based on the natural distribution of data and a hunch about how the proportion of spam might change throughout the day.
- We might expect that spammers operate in the late hours of the night or they may operate during the day, when people are likely to check their e-mail.
- This said, to capture these trends, we could have just as easily used three bins or twelve.

- **Too few bins** can result in important trends being obscured.
- **Too many bins** can result in small counts in the Naive Bayes frequency table.
- Which can increase the algorithm's sensitivity to noisy data.

PYQ DISCUSSION

Third Semester MCA (2 Year) Degree Examination December 2021

- What are the strengths and weaknesses of K-NN algorithm
- Based on the survey conducted in an institution the students are classified based on the 2 attributes academic excellence and other achievements. Consider the data set given. Find the classification of a student with value of X is 5 and Y is 7 based on the data of trained samples using KNN algorithm. Choose $k = 3$

X [Academic Excellence]	Y [Activities]	Z [Classification]
8	6	Outstanding
5	6	Good
7	3	Good
6	9	Outstanding

- Consider a training data set consisting of the fauna of the world. Each unit has 3 features named “Swim”, “Fly” and “Crawl”. Let the possible values of these features be as follows:

Swim - Fast, Slow, No

Fly - Long, Short, Rarely, No

Crawl - Yes, No

For simplicity, each unit is classified as “Animal”, “Bird” or “Fish”. Let the training data set be as in the table below . Use naive Bayes algorithm to classify a particular species if its features are (Slow, Rarely, No)

Sl. No.	Swim	Fly	Crawl	Class
1	Fast	No	No	Fish
2	Fast	No	Yes	Animal
3	Slow	No	No	Animal
4	Fast	No	No	Animal
5	No	Short	No	Bird
6	No	Short	No	Bird
7	No	Rarely	No	Animal
8	Slow	No	Yes	Animal
9	Slow	No	No	Fish
10	Slow	No	Yes	Fish
11	No	Long	No	Bird
12	Fast	No	No	Bird

Third Semester MCA (Two Year) Regular and Supplementary Examination December 2022

- Explain methods to prepare data for use with k-NN.
- Consider the given dataset. Apply Naïve Bayes algorithm and predict that if a fruit has following properties, then which fruit it is. Fruit = (Yellow, Sweet, Long)

Fruit	Yellow	Sweet	Long	Total
Mango	350	450	0	650
Banana	400	300	350	400
Others	50	850	400	150
Total	800	850	400	1200

- Given the following dataset. Identify the T-Shirt Size of Tom having height 161 cm and weight 61kg using k-NN algorithm.

(Choose k as 3)

Height (cm)	Weight (kg)	T-Shirt Size
158	58	Medium
158	59	Medium
158	63	Medium
160	59	Medium
160	60	Medium
163	60	Medium
163	61	Medium
160	64	Large
163	64	Large
165	61	Large
165	62	Large
165	65	Large
168	62	Large
168	63	Large
168	66	Large
170	63	Large
170	64	Large
170	68	Large

Third Semester MCA (Two Years) Degree (R, S) Examination December/January 2023-24

- Illustrate the learning process by machine.
- Can Naive Bayes algorithm handle Numerical continuous variables? Justify your answer.
- Consider the dataset given below. Using k-NN algorithm, predict the class label for the new instance with height=172 cm and weight =57 kg. Choose k=1 and k=3

Height (cm)	Weight (kg)	Class
167	51	Underweight
182	62	Normal
176	69	Normal
173	64	Normal
172	65	Normal
174	56	Underweight
169	58	Normal
173	57	Normal
170	55	Normal
169	53	Underweight

Given a training dataset. Predict the Species type of new instance with Colour=Brown, Legs=2, Height=Tall, Smelly=No using Naive bayes classifier

Colour	Legs	Height	Smelly	Species
White	3	Short	Yes	M
Brown	2	Tall	No	M
Brown	3	Short	Yes	M
White	3	Short	Yes	M
Brown	2	Short	No	H
White	2	Tall	No	H
White	2	Tall	No	H
White	2	Tall	Yes	H

Third Semester MCA (Two Year) Degree (R,S) Examination December 2024

- Explain the role of Laplace estimator in Bayesian classification.
- Briefly discuss different types of learning algorithms.

With the given data, Use k-NN algorithm to determine the Target attribute for a new instance with $X = 5$ and $Y = 3$. (Choose k as 3)

X	Y	TARGET
2	3	Class1
3	3	Class1
2	8	Class2
4	1	Class1
5	8	Class2
6	7	Class2

- Consider the training data of 10 samples in the given table where ‘Play’ is a class attribute. Use Bayesian classifier to predict whether there will be a play if it is a rainy day with mild temperature, Normal humidity and Strong wind.

Day	Outlook	Temperature	Humidity	Wind	PLAY
1	Sunny	Hot	High	Weak	No
2	Sunny	Cool	Normal	Weak	Yes
3	Overcast	Hot	High	Weak	Yes
4	Sunny	Mild	High	Weak	No
5	Sunny	Mild	Normal	Strong	Yes
6	Rain	Mild	High	Weak	Yes
7	Rain	Cool	Normal	Weak	Yes
8	Rain	Cool	Normal	Strong	No
9	Sunny	Hot	High	Strong	No
10	Overcast	Mild	High	Strong	Yes

MCA (Two Years) 3rd Semester (S,FE) Examination June 2025

- Explain disadvantages of K-NN classifier.
- What is meant by “learning” in the context of machine learning?
- Consider the dataset given below. Using k-NN algorithm, predict the class label for the new instance with brightness=20 and saturation =35. Choose k=1 and k=3.

Brightness	Saturation	Class
40	20	Red
50	50	Blue
60	90	Blue
10	25	Red
70	70	Blue
60	10	Red
25	80	Blue

14

Given a training dataset. Predict the class of a new patient with the symptoms Fever: Yes, Cough: No, Body Ache: Yes, Fatigue: No, using Naive Bayes classifier.

Patient	Fever	Cough	Body Ache	Fatigue	Disease
1	Yes	Yes	No	Yes	Disease
2	No	Yes	Yes	No	No Disease
3	Yes	No	Yes	Yes	Disease
4	Yes	Yes	Yes	No	No Disease
5	No	Yes	No	No	No Disease
6	Yes	Yes	Yes	Yes	Disease
7	Yes	No	No	Yes	Disease
8	No	No	Yes	No	No Disease
9	Yes	Yes	No	Yes	Disease
10	No	No	No	No	No Disease

MCA (Two Years) Degree / PhD 3rd Semester (R,S) Examination November 2025

- Describe why nearest neighbour classifiers are considered "lazy" learners
- Explain overfitting in Machine Learning. Explain the techniques used to limit overfitting.

13 By using KNN algorithm find in which class new data entry falls.

Brightness Saturation Class

40	20	Red
50	50	Blue
60	90	Blue
10	25	Red
70	70	Blue
60	10	Red
25	80	Blue.

The new data with Brightness is 20 and Saturation is 36. Find the Class.

OR

14 What are Overfitting and Under fitting? Explain the procedures to reduce under fitting.

MCA (Two Year) Degree S3 (S,FE) Examinations June 2026

- Why is the choice of k important in k -NN?
- What is the Laplace estimator? Discuss its importance in handling zero probability issues in Naïve Bayes.
- Explain the significance of Laplace estimator in Bayesian classification. Explain the different ways to prepare numeric features in naive Bayes algorithm.

You are given the following dataset with two features (Feature 1 and Feature 2) and corresponding labels (Class A and Class B):

Feature 1	Feature 2	Label
1	2	A
2	3	A
3	3	A
6	5	B
7	7	B
8	6	B

Use K-Nearest Neighbors (KNN) with $K = 3$ to classify the following test point: (5, 5).